

## User Interface



Q. Is this tomato ripe?



Yes (Ripe)



No (Not Ripe)



Human operator

## Robot & Environment



## Knowledge-based Planning System

### Knowledge Base

- facts about objects
- robot states / actions
- task knowledge



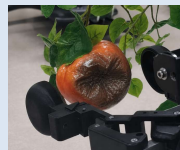
### POMCP

Select action

### Belief over Possible Worlds (Uncertainty)

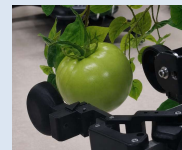


$s_1: 0.6$



$s_2: 0.3$

...



$s_N: 0.1$

**Belief Update & Uncertainty Measure**  
(from observations / feedback)

$conf > \tau$

Select Informative  
Query

Update  
Knowledge Base

Feedback

Query

Observation

Action